Course Six

The Nuts and Bolts of Machine Learning

Instructions

Use this PACE strategy document to record decisions and reflections as you work through the end-of-course project. As a reminder, this document is a resource that you can reference in the future and a guide to help consider responses and reflections posed at various points throughout projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- Complete the questions in the Course 6 PACE strategy document
- Answer the questions in the Jupyter notebook project file
- Build a machine learning model
- Create an executive summary for team members and other stakeholders

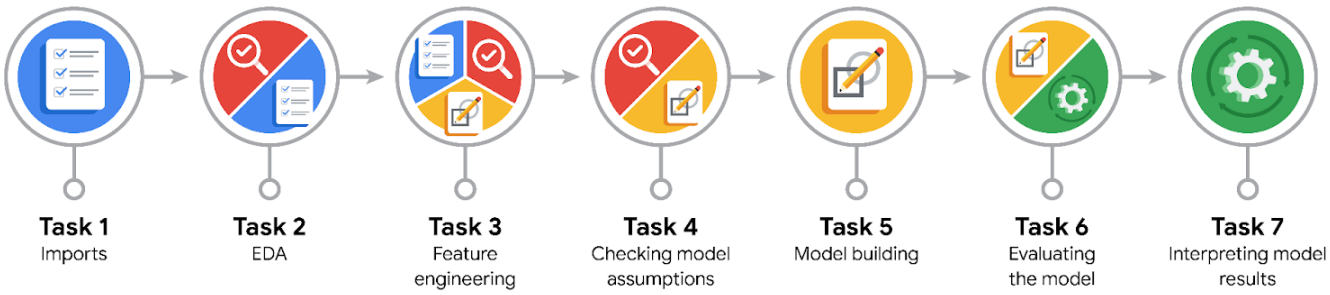
Relevant Interview Questions

Completing the end-of-course project will empower you to respond to the following interview topics:

- What kinds of business problems would be best addressed by supervised learning models?
- What requirements are needed to create effective supervised learning models?
- What does machine learning mean to you?
- How would you explain what machine learning algorithms do to a teammate who is new to the concept?
- How does gradient boosting work?

Reference Guide:

This project has seven tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- What are you trying to solve or accomplish?

  To predict if a customer is a generous tipper, that being a customer who tips 20% or more.

- Who are your external stakeholders that I will be presenting for this project?

  Juliana Soto (Finance and Administration Department Head) and Titus Nelson (Operations Manager).

- What resources do you find yourself using as you complete this stage?

  NumPy, pandas, seaborn, matplotlib, sci-kit learn, pickle and xgboost.

- Do you have any ethical considerations at this stage?

  The ethical considerations that I have at this stage is that of the effects when the model makes an error, such as a false negative or false positive. If the model predicts a false negative, this is where the model predicts a tip less than 20% but the customer ends up giving more. If the model predicts a false positive, the model is predicting that a customer will tip 20% or more but the customer ends up not giving one. Both of these errors have ethical implications for both the driver and the customer, such as the driver not accepting the ride or the driver's revenue being decreased.

- Is my data reliable?

Yes my data is reliable as it is primary data that comes from NYC TLC.

- What data do I need/would like to see in a perfect world to answer this question?

The data that I would like to see in order to answer this question is past tipping behaviour for each customer such as the number of tips a customer has given and the average tip amount.

- What data do I have/can I get?

The data I have is the NYC TLC Yellow Taxi Trip Data from 2017 and the data I can get is the predicted fare amount along with the mean duration and mean distance from the multiple linear regression analysis.

- What metric should I use to evaluate success of my business/organizational objective? Why?

The evaluation metric that should be used is the f1-score, as this is the balance of precision and recall. This is the best metric to use for our business objective as both false negatives and false positives are detrimental and so we want to minimise both while balancing the precision and recall.

PACE: Analyze Stage

- Revisit "What am I trying to solve?" Does it still work? Does the plan need revising?

It does still work and the plan does not need revising as we need to feature engineer some new variables.

- Does the data break the assumptions of the model? Is that ok, or unacceptable?

The data has extreme values/outliers and these don't break the assumptions of the model, as ensemble decision trees methods such as Random Forests and XGBoosts are robust to outliers. Therefore the data does not break the assumptions of the model.

- Why did you select the X variables you did?

I selected the X variables due to their relevance to classifying a generous tipper of 20% or more, with these being majority of the engineered features, trip metrics, predicted fare and passenger metrics. For example, some of the variables chosen were RatecodeID, passenger_count, VendorID, mean_duration/distance, generous, am/pm_rush and predicted_fare. The variables that were not selected were irrelevant to the objective such as mta_tax and improvement_surcharge.

- What are some purposes of EDA before constructing a model?

Some of the purposes of EDA before constructing a model is to better understand the data and prepare it for modelling, such as through structuring, cleaning or feature engineering. Through feature engineering we can make new features that can give more predictive signal in the data for the model.

- What has the EDA told you?

The EDA has told me that we engineered new features such as tip_percent, generous, day, month, am/pm_rush, daytime and nighttime. Additionally, we selected the X variables for our model by dropping the variables irrelevant to our business objective and encoded our categorical variables into dummy variables. Finally, we know from our EDA that there is a balance in the class balance of our target variable generous, with a little over half of the customers being a generous tipper.

- What resources do you find yourself using as you complete this stage?

Pandas and datetime

PACE: Construct Stage

- Do I notice anything odd? Is it a problem? Can it be fixed? If so, how?

No I do not notice anything odd.

- Which independent variables did you choose for the model, and why?

The independent variables I chose for the model were: VendorID, passenger_count, RatecodeID, PULocationID, DOLocationID, mean_duration, mean_distance, predicted_fare, day, am_rush, daytime, pm_rush, nighttime and month. I chose these because of their significant relevance to the business objective of predicting a generous tipper, as well as the engineered features being able to generate more predictive signal for the model.

- How well does your model fit the data? What is my model's validation score?

The model fits the data extremely well, with the validation scores of precision of 69%, recall of 83%, f1-score of 75% and an overall accuracy of 71% on the training data. In comparison, my models score on the unseen test set was a precision of 68%, a recall of 82%, a f1-score of 74% and an overall accuracy of 70%.

- Can you improve it? Is there anything you would change about the model?

Yes the model can be improved by engineering more features from the data to produce more predictive signal. Additional data can be incorporated with data on past tipping behaviours of the customers which can improve model performance.

- What resources do you find yourself using as you complete this stage?

Pandas, NumPy, Sci-kit learn, pickle, xgboost, matplotlib and seaborn.

PACE: Execute Stage

- What key insights emerged from your model(s)? Can you explain my model?

The key insights that emerged from my model is that the VendorID 2 was the most influential variable to the models predictions. The VendorID of 2 is the TPEP provider of VeriFone Inc. and from the feature importance plot of the most influential in predicting a generous tipper. Other influential features are passenger_count, mean_duration, predicted_fare and mean_distance, thereby indicating that the characteristics of the journey and its fare size influence the models predictions.

- What are the criteria for model selection?

The criteria for model selection is transparency and explainability, therefore the Random Forest model was chosen as the champion model.

- Does my model make sense? Are my final results acceptable?

Yes my model makes sense and the final results are acceptable as the model generalises well onto unseen data and its predictions are transparent and explainable.

- Do you think your model could be improved? Why or why not? How?

Yes the model can be improved either by more feature engineering or additional data, such as data on customers past tipping behaviour such as stating the number of tips given, average tip amount or a binary variable that states whether the historical tips of the customer has been equal to or more than 20%.

- Were there any features that were not important at all? What if you take them out?

Yes there were features that played little importance at all such as am/pm_rush, nighttime, daytime, Majority of the PU/DOLocationID's, day and month. If these were either used for further feature engineering then taken out or taken out entirely and replaced with more predictive features than this could improve the model.

- What business/organizational recommendations do you propose based on the models built?

The model's results indicated that the tipping behaviour of customers is driven primarily by the vendor, fare size and trip characteristics, such as the amount of passengers and predicted fare amount. Therefore, NYC TLC should focus on improving the service across the vendors and can introduce targeted tipping prompts for high value trips and peak time trips. Additionally, NYC TLC can introduce pricing or promotional strategies for group travel as these trips appear to have a higher tipping potential. Also the social dynamics of groups encourage more generous tipping via social pressure/shared costs and thereby is a relevant segment for revenue optimisation. These recommendations can help increase the tipping rates and overall revenue.

- Given what you know about the data and the models you were using, what other questions could you address for the team?

The other questions that I can address for the team are: is the model ethical? Can the models results be explained? How well does the model generalise to unseen data? How can the models results be used to benefit taxi drivers of TLC NYC?

- What resources do you find yourself using as you complete this stage?

Pandas, matplotlib and scikit-learn.

- Is my model ethical?

Yes my model is ethical as it balances precision and recall and balances the amount of false positives and negatives.

- When my model makes a mistake, what is happening? How does that translate to my use case?

When my model makes a mistake it is either making a false positive or negative, with both of them being detrimental to the business objective. For example, a false positive is when the model predicts a tip of 20% or more but the customer doesn't give one. This leads to the driver earning less revenue and believing they selected a journey probable of a generous tip without one which can lead to dissatisfaction with NYC TLC itself. Additionally, if the model predicts a false negative which is when the model predicts a tip of less than 20% but the customer gives more, then this can lead to the driver not accepting the journey due to thinking the customer won't be giving a generous enough tip.